



Tertiary Infotech Academy Pte Ltd

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LAB 09 INSTRUCTIONS

For

Diagnose DNS failure and delay

Course Code: C1123

Conducted by

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Version 5.1

Lab 09 — Diagnose DNS failure and delay

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 09 — DNS Traffic Analysis

Learning outcome: LO3 — Interpret DNS, ARP, IPv4, ICMP and UDP evidence.

Time: about 60 minutes

Version: v5.1 · 5 October 2026

Scenario

Users report that some names fail and others resolve slowly. You separate a missing name from a slow resolver using the DNS evidence.

Goal

Transaction ID plus addresses and ports pair a query with its response.

What you will produce

A DNS observations sheet with measured intervals.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/dns-observations.csv	DNS observations sheet you complete

Lab at a glance

1. Pair DNS queries and responses by ID
2. Explain the NXDOMAIN response
3. Measure the 0.8 s slow transaction
4. Record observations and next steps

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `dns` and add `dns.time` as a column from a response packet. Pair IDs 101, 102 and 103.

3. Apply `dns.flags.rcode == 3`. Expand DNS flags and name. Record missing `example.test` and explain NXDOMAIN.

4. Apply `dns.id == 103`. Measure the query-response interval: 0.800 seconds. Compare with the two 0.020 second transactions.

5. Complete `assets/dns-observations.csv` with name, transaction ID, rcode, elapsed seconds and next investigation. Do not label this measured delay as a DNS timeout.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

dns.flags.rcode == 3					
No.	Time	Source	Destination	Protocol	Info
6	0.070000	192.0.2.53	192.0.2.10	DNS	Standard query response 0x0066 No such name A mis...
branch-office.pcap Displayed: 1					

Lab 09 expected evidence

Test it

The supplied `dns.flags.rcode == 3` expression matches 1 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching `lab-tls.keys` file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y "dns.flags.response == 1" -T  
fields -e dns.id -e dns.qry.name -e dns.flags.rcode -e dns.time
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

DNS lab — compare A, NS and MX lookups in the DNS trace from the Kurose & Ross labs.

Source: J.F. Kurose and K.W. Ross, Wireshark Labs

(gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.